IF A $[[14, 3, 5]]$ STABILIZER CODE EXISTS, ITS MONOMIAL AUTOMORPHISM GROUP HAS ORDER $2^a3^b5^c$: A CERTIFICATE-BACKED EXCLUSION OF THE PRIMES 7, 11 AND 13

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Abstract. Whether a $[[14, 3, 5]]$ qubit stabilizer code exists is open, and this note does not resolve it. The $[[14, 3]]$ entry of Grassl's online tables records lower bound 4 (construction dated June 30, 2005) and upper bound 5 (page retrieved August 5, 2026) [Gra], and Ball, Centelles and Huber pose the existence question explicitly [BCH20]. What is proved here is a structure theorem: if a $[[14, 3, 5]]$ exists, its group of monomial automorphisms — local Clifford $\times$ qubit permutation, equivalently monomial symplectic maps on $\mathbb{F}_2^{28}$ — contains no element of order divisible by 7, 11 or 13, and therefore has order $2^a3^b5^c$. The proof is an exhaustive, certificate-backed elimination of every conjugacy class of monomial maps of order 7, 11 and 13: three of the four prime classes are closed by complete enumeration of the invariant isotropic 11-dimensional subspaces, every candidate certified to have $d \leq 4$ by an exhibited low-weight vector (1,298,700 candidates for the fixed-point-free order-7 class, 8,415 for order 11, none for order 13); the fourth, order 7 with seven fixed qubits, is closed by an exact-rational linear-programming certificate (Krawtchouk positivity: every additive $(7, 2^9)$ code has at least 52 nonzero vectors of symplectic weight at most 4) together with two short counting arguments. The composite-order and locally twisted classes are enumerated as well, as redundant confirmation. The corpus was re-verified adversarially with independently written code, and the LP and branch certificates replay with the stock Python interpreter.

The CSS case is settled by Koh et al. [Koh26], whose exhaustive enumeration of CSS codes through $n = 14$ shows that the maximal distance of a CSS $[[14, 3]]$ code is 4; we cite this and claim nothing there. The open territory for $[[14, 3, 5]]$ is thus genuinely non-CSS, with monomial symmetries confined to $\{2, 3, 5\}$-groups. Negative search data is recorded as boundary information, not as evidence.

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Computation and authorship. All reductions, enumerations, certificate designs and searches in this work were produced by **Claude Fable 5** (Anthropic), directed by the author, on a single Apple M4 laptop (10 cores, 16 GB). The enumeration corpus was then re-verified adversarially by a separate agent instance using independently written checker code that shares nothing with the generating pipeline: every enumerated candidate of every nonzero symmetry class was re-checked, and the linear-programming certificate of Lemma 4.2 was replayed in exact rational arithmetic. This is a factual methods statement, and it is part of the point: the artifacts are designed so that the provenance of the *search* is irrelevant to the validity of the *result*. External tools used by the pipeline only: a C compiler for the distance checker, SymPy for one polynomial factorization; the closure certificates themselves replay with the stock Python interpreter (§7).

Status and provenance record. The problem statement was verified against the primary source (codetables.de, retrieved August 5, 2026, raw HTML archived with the artifacts), and a novelty sweep was completed the same day: no published existence or nonexistence result for $[[14, 3, 5]]$ was found beyond the CSS case of [Koh26], and no published automorphism restriction for it was found. A negative cannot be proved this way; we record the scope of the search rather than a claim. *Draft of August 5, 2026.*

1. Introduction

1.1. **The problem.** Let $[[n, k, d]]$ denote a binary (GF(4)-additive) quantum stabilizer code on n qubits with k logical qubits and minimum distance d [Got97, CRSS98]. The entry $[[14, 3]]$ of Grassl's online tables [Gra] records lower bound 4 and upper bound 5; the $[[14, 3, 4]]$ construction there is dated June 30, 2005 (page last changed June 10, 2024, retrieved August 5, 2026). Both neighbors exist: $[[15, 3, 5]]$ and $[[14, 2, 5]]$ are settled constructions in the same tables, so $[[14, 3, 5]]$ is exactly the missing corner. Ball, Centelles and Huber raise its existence as Research Problem 1 of [BCH20]. For context on nearby exhaustive work: Bierbrauer, Fears, Marcugini and Pambianco proved $[[13, 5, 4]]$ does not exist [BFMP11]; Cross and Vandeth enumerate all general stabilizer codes through $n = 9$ [CV25]; and Koh et al. enumerate all CSS codes through $n = 14$ [Koh26]. On the quantum side, automorphism groups of stabilizer codes have been studied in their own right — Hao [Hao21] proves nonexistence results for stabilizer codes with highly transitive automorphism groups — but that work contains nothing at length 14.

The existence of $[[14, 3, 5]]$ **remains open.** This note proves a restriction on what such a code could look like, and ships the machine-checkable certificates behind it. It does not resolve the question.

1.2. **The result.** Call a linear map on $\mathbb{F}_2^{28}$ *monomial* if it is a qubit permutation composed with an invertible local symplectic map on each qubit's (X, Z)-plane — in operator language, a local Clifford circuit times a qubit permutation, modulo Paulis; in GF(4) language, a monomial semilinear map. These maps form the group $\mathcal{M} \cong S_3 \wr S_{14}$ of order $6^{14} \cdot 14!$, whose prime divisors are exactly $2, 3, 5, 7, 11, 13$. A *monomial automorphism* of a stabilizer code with stabilizer subspace $S \subset \mathbb{F}_2^{28}$ is a monomial map with $g(S) = S$.

Theorem 1.1 (main; Theorem 3.1 below)**.** *No* $[[14, 3]]$ *stabilizer code of minimum distance* ≥ 5 *admits a monomial automorphism of order* 7*,* 11 *or* 13*.*

Corollary 1.2. *If a* $[[14, 3, 5]]$ *code exists, no element of its monomial automorphism group has order divisible by* 7*,* 11 *or* 13 *(Cauchy's theorem applied to a cyclic subgroup), so the group order is of the form* $2^a 3^b 5^c$*.*

Two by-products are worth stating at their own strength. First, no $[[14, 3]]$ stabilizer code of *any* distance has a monomial automorphism of order 13: the invariant-subspace dimension arithmetic is already empty (Proposition 3.2). Second, every $[[14, 3, 5]]$ would be a hyperplane (index-2 subgroup) of a $[[14, 2, 5]]$ code (Proposition 6.1), which turns hyperplane sweeps of known $[[14, 2, 5]]$ codes into complete searches of their neighborhoods; the two such sweeps performed here were negative (§6).

1.3. **What is not claimed.** We do not claim that $[[14, 3, 5]]$ does not exist, and nothing here is evidence either way about codes with trivial monomial symmetry, which one expects to be the bulk of the search space. We do not claim the CSS case: that is Koh et al.'s [Koh26], whose Table VI shows the maximal distance of a CSS $[[14, 3]]$ code (pure or impure) is 4; our own weaker pure-CSS observation appears only as Remark 6.2, crediting them. We do not claim novelty for the prescribed-automorphism technique, which is classical in coding theory (Huffman [Huf98]; see also [CRSS98]); what appears to be new, per the dated sweep in the front matter, is its certified application to this open entry. The very shape of the theorem — pinning the automorphism group of a putative extremal code down to a small set of orders by excluding primes class by class — is also classical: it is the shape of the long program on the putative binary doubly-even self-dual $[72, 36, 16]$ code, opened by Conway and Pless in

1982 and continued by many authors (the surviving group orders there are now 1, 3 and 5, by work of Borello and others); see Huffman [Huf98] for the early history. The constructive searches of §6 are boundary data, not evidence: a rugged objective landscape says nothing about existence. Finally, the theorem concerns *monomial* automorphisms only; equivalences that entangle qubits are not addressed.

2. Setting

We use the binary symplectic picture [CRSS98, Got97]. A vector $v \in \mathbb{F}_2^{28}$ is a pair (a, b) of 14-bit words (X- and Z-parts); the symplectic form is $\omega(u, v) = |u_a \wedge v_b| + |u_b \wedge v_a| \bmod 2$, and the symplectic weight $\mathrm{wt}(v) = |a \vee b|$ is the number of qubits touched. A [[14, 3]] stabilizer code is an isotropic subspace $S \subset \mathbb{F}_2^{28}$ of dimension 11; then $\dim S^\perp = 17$ and the minimum distance is

$$d \;=\; \min\{\mathrm{wt}(v) : v \in S^\perp \setminus S\},$$

a minimum over $2^{17} - 2^{11} = 129{,}024$ vectors. Monomial maps preserve ω and wt, so if g is monomial and $g(S) = S'$ then S' is a code with the same parameters; in particular, closing a conjugacy class of monomial maps for one representative closes it for all.

Lemma 2.1 (prime-order normal form)**.** *Let $g \in \mathcal{M}$ have prime order $p \in \{7, 11, 13\}$. Then g is conjugate in $\mathcal{M}$ to a pure qubit permutation of cycle type $(7, 7)$ or $(7, 1^7)$ if $p = 7$, type $(11, 1^3)$ if $p = 11$, and type $(13, 1)$ if $p = 13$.*

Proof. Write $g = (t_1, \ldots, t_{14}; \pi)$ with local parts $t_i \in \mathrm{Sp}(2, \mathbb{F}_2) \cong S_3$ and $\pi \in S_{14}$. Since S_3^{14} has no element of order p, $\pi \neq 1$; $\pi^p = 1$ forces the stated cycle types. At a fixed point i of π, $g^p = 1$ gives $t_i^p = 1$, and $\gcd(p, 6) = 1$ forces $t_i = 1$. Along a p-cycle, the p-th power acts by conjugates of the cycle product $T = t_{i_p} \cdots t_{i_1}$, so $T = 1$; and a cycle with trivial product is conjugate, by a coordinatewise gauge u_j chosen recursively along the cycle, to the untwisted cycle. □

So the theorem reduces to four classes of pure permutations. The certified record (§3–4) in fact closes more: all monomial classes with a 14-cycle permutation part (orders 14, 28, 42) and all locally twisted classes over the $(7, 7)$ cycle type (orders 14, 21, 42; no order-28 class exists there, S_3 having no element of order 4). By Lemma 2.1 these composite-order closures are logically redundant — a power of any such element lands in a prime class — and we report them as confirmation.

3. The enumerated classes

Theorem 3.1. *No [[14, 3]] stabilizer code with $d \geq 5$ admits a monomial automorphism of order 7, 11 or 13.*

The proof occupies this section and the next. For a fixed representative σ, the σ-invariant isotropic 11-dimensional subspaces are enumerated exhaustively, and each candidate is screened by a distance scan with early abort at weight ≤ 4: the scan certifies $d \leq 4$ by exhibiting a weight-≤ 4 vector of $S^\perp \setminus S$, and any survivor would have received its exact distance by full enumeration of $S^\perp \setminus S$ (129,024 vectors). No candidate survived. Table 1 is the complete certified record; every nonempty row traces to a file hashed in §7 (the empty `c13` class leaves no output file; its emptiness replays via `gen_generic.py c13`).

class (representative σ)	order	candidates	outcome
14-cycle, untwisted (`cyclic`; `swapshift` re-run)	14	1,260	all $d \le 4$
14-cycle, involution twist (`shift28`)	28	4	all $d \le 4$
14-cycle, order-3 twist (`s3shift`, `s3shift2`)	42	0	class empty
$(7,7)$, twist $(1,1)$ (`qc7`, jobs `dp0`–`dp3`)	7	1,298,700	all $d \le 4$
$(7,7)$, twist $(1,s)$ (`qc7_1swap`)	14	4,764	all $d \le 4$
$(7,7)$, twist (s,s) (`qc7swap`)	14	540	all $d \le 4$
$(7,7)$, twist (s,s') (`qc7_swap_s2`)	14	540	all $d \le 4$
$(7,7)$, twists with an order-3 part (4 classes)	$21, 42$	0	classes empty
11-cycle $+\,3$ fixed (`c11`)	11	8,415	all $d \le 4$
13-cycle $+\,1$ fixed (`c13`)	13	0	class empty
7-cycle $+\,7$ fixed	7	—	§4

TABLE 1. The certified class closures: 1,314,223 distinct enumerated candidates (1,315,483 distance checks in all, the `swapshift` re-run of the cyclic class contributing a further 1,260), each certified $d \le 4$ by the early-abort scan described above, none reaching $d = 5$. Twists over $(7,7)$ are classified by the pair of cycle products in S_3 up to simultaneous conjugacy and block swap (s, s' involutions in different positions); over a 14-cycle, by the single cycle product.

3.1. **Enumeration method.** Each representative σ is an explicit linear map on $\mathbb{F}_2^{28}$ preserving ω. The invariant subspaces are enumerated through the primary (Fitting) decomposition of $\mathbb{F}_2^{28}$ as an $\mathbb{F}_2[\sigma]$-module — the classical prescribed-automorphism method [Huf98] — with complete submodule enumeration per primary component and isotropy imposed through the pairing of components (the reciprocal pairing swaps the two degree-3 factors of x^7+1, so only cross-orthogonality constraints survive there). Two structural examples:

Cyclic. For the qubit 14-cycle, $x^{14}+1 = \big((x+1)(x^3+x+1)(x^3+x^2+1)\big)^2$ over $\mathbb{F}_2$; the invariant isotropic 11-dimensional subspaces number exactly 1,260 (7 isotropic choices in the $(x+1)$-component times 180 orthogonal pairs in the two cubic components).

QC-7. For two disjoint 7-cycles, $x^7+1 = (x+1)pq$ is squarefree with p, q cubic; an invariant subspace is $U_0 \oplus U_p \oplus U_q$ with U_0 a Lagrangian plane in the 4-dimensional fixed component (15 choices) and (U_p, U_q) an orthogonal pair of K-subspaces, $K = \mathbb{F}_8$ ($585+42{,}705+42{,}705+585 = 86{,}580$ pairs), for $15 \times 86{,}580 = 1{,}298{,}700$ candidates.

Every emitted candidate is re-checked for rank 11 and isotropy before its distance is computed (“0 bad” in every batch log). The batch screener also has a positive control: run with generator count 12 on the reference $[[14,2,5]]$ matrix, it flags the code as a $d \ge 5$ hit. (The control is a separate `batch 12` run — a 12-generator matrix cannot ride the 11-generator stream, where it would be counted “bad” — and its console output was not archived; it replays in seconds.)

3.2. **Order 13.**

Proposition 3.2. *No* $[[14,3]]$ *stabilizer code, of any distance, has a monomial automorphism of order* 13.

Proof. By Lemma 2.1 the automorphism may be taken to be a pure permutation σ: a 13-cycle plus a fixed qubit. Then the trivial $\mathbb{F}_2[\sigma]$-component of $\mathbb{F}_2^{28}$ has dimension 4 and Φ_{13} is irreducible of degree 12 over $\mathbb{F}_2$ (the order of 2 mod 13 is 12), so any invariant subspace has dimension $d_0 + 12m$ with $d_0 \le 4$. No such value equals 11. The generic enumerator confirms: zero invariant 11-dimensional subspaces (Table 1). ☐

The order-11 class is not empty — the Φ_{11}-part is a line over $\mathbb{F}_{2^{10}}$, giving 33 isotropic lines $\times\, 255$ choices of fixed-component part $= 8{,}415$ candidates — and all 8,415 fail. By Lemma 2.1, orders 11 and 13 are thereby fully monomially closed.

4. Order 7 with fixed qubits

Let σ be the permutation acting as a 7-cycle on qubits 0..6 and fixing qubits 7..13, and suppose S is a σ-invariant [[14, 3, 5]] stabilizer subspace. Decompose $S = U_0 \oplus U_p \oplus U_q$ as above, now with the trivial component of $\mathbb{F}_2^{28}$ of dimension 16 (the 14 fixed-qubit coordinates plus the two block averages), a symplectic space. Write $d_0 = \dim U_0$; then $d_0 + 3(d_p + d_q) = 11$, so $d_0 \in \{2, 5, 8, 11\}$. Each branch is impossible:

Lemma 4.1 ($d_0 = 11$)**.** *U_0 would be an isotropic subspace of dimension 11 in a 16-dimensional symplectic space, whose maximal isotropic dimension is 8.*

Lemma 4.2 (LP certificate; `lemmaC_certificate.py`)**.** *Every additive $(7, 2^9)$ code over GF(4) — equivalently, every 9-dimensional $\mathbb{F}_2$-subspace of the 14-dimensional symplectic space of 7 qubits — has at least 52 nonzero vectors of symplectic weight at most 4.*

Proof. Delsarte-style linear programming [Del73] with the quaternary Krawtchouk polynomials K_j and the MacWilliams nonnegativity of the trace-dual weight distribution of an additive GF(4) code [CRSS98]. With $\mu = \frac{5}{32}$ and $\lambda = (0, \frac{27}{256}, 0, 0, 0, \frac{1}{256}, 0, \frac{1}{384}) \geq 0$, one checks exactly that $e(w) := [w \leq 4] - \mu - \sum_j \lambda_j K_j(w) \geq 0$ for $w = 1, \ldots, 7$, whence for any $(7, 2^9)$ additive code, $A_1 + \cdots + A_4 \geq 511\mu - \sum_j \lambda_j K_j(0) = 52$. The shipped certificate performs every step in exact rational arithmetic (Python `fractions`); it was re-verified independently. (An empirical floor of 104 over 2,000 random subspaces suggests the constant is not tight; harmless.) □

Closure of $d_0 \in \{2, 5\}$. Let $V_{\text{fix}} \cong \mathbb{F}_2^{14}$ be the span of the fixed-qubit coordinates and let $W = \{f \in V_{\text{fix}} : f \perp_\omega U_0\}$. Since V_{fix} is ω-orthogonal to the block components and to the block-average plane, $W \subseteq S^\perp$, and $\dim W \geq 14 - d_0 \geq 9$. Moreover $W \cap S = U_0 \cap V_{\text{fix}}$ has at most $2^{d_0} - 1 \leq 31$ nonzero vectors, and every vector of $W \setminus S$ lies in $S^\perp \setminus S$, hence has weight ≥ 5. But W contains a 9-dimensional subspace supported on 7 qubits, which by Lemma 4.2 has at least 52 nonzero vectors of weight ≤ 4: contradiction. □

Closure of $d_0 = 8$. Here $d_p + d_q = 1$; by the $p \leftrightarrow q$ relabeling symmetry take $(d_p, d_q) = (1, 0)$. The block-supported subspace $M = V_p \oplus (U_p^\perp \cap V_q)$ has dimension 9 and lies in $S^\perp$; its weight-≤ 4 vectors must lie in the block-only part of S, contained in a subspace B_S of dimension ≤ 5 (at most 31 nonzero vectors). Direct enumeration over all 9 possible K-lines U_p counts 112–140 nonzero weight-≤ 4 vectors in M (`order7fixed_full.txt`): contradiction. (M is itself 9-dimensional on 7 qubits, so Lemma 4.2 independently forces ≥ 52: the branch dies twice.) □

Proof of Theorem 3.1. By Lemma 2.1 it suffices to treat pure permutations of the four cycle types. Types $(7, 7)$, $(11, 1^3)$ and $(13, 1)$ are closed by the exhaustive enumerations of Table 1; type $(7, 1^7)$ by Lemmas 4.1 and 4.2 with the branch arguments above. The composite-order rows of Table 1 are consistent redundant confirmation. □

5. Verification

The corpus is guarded at four levels.

Controls. The C distance checker reproduces $d = 4$ on the codetables [[14, 3, 4]] matrix and $d = 5$ on the codetables [[14, 2, 5]] matrix before any enumeration is trusted; batch mode carries the positive control described in §3.

Independent verifier. A standard-library-only Python verifier, written separately, sharing no code with the search tools and using a different algorithm (explicit construction of the full 2^{11}-element group and 2^{17}-element normalizer), agrees with the C checker on the codetables $[[14, 3, 4]]$ reference code (both find $d = 4$; the verifier reports FAIL by design, since $4 < 5$) and on a random sample of 8 enumerated cyclic candidates, with exact distance agreement in all cases. (The verifier accepts only 11-generator $[[14, 3]]$ inputs, so the 12-generator $[[14, 2, 5]]$ reference is outside its scope; the sample's console output was not archived.)

Structural cross-checks. The cyclic class was enumerated by two structurally different programs (CRT-idempotent construction; generic kernel-of-matrix method): identical count (1,260) and identical component submodule statistics. The QC-7 class likewise: the specialized Grassmannian generator emits exactly 1,298,700 candidates, the generic tool independently reproduces that count (a count-only cross-check: candidates constructed and isotropy-checked, not re-distance-checked), and the orthogonal-pair counts match the Gaussian-binomial predictions over $\mathbb{F}_8$ exactly ($585 \times 73 = 42{,}705$; $4{,}745 \times 9 = 42{,}705$).

Adversarial re-verification. After the corpus was frozen, a separate agent instance with independently written checker code re-ran the verification: every enumerated candidate of every nonzero symmetry class in Table 1 was re-checked, and Lemma 4.2 was replayed in exact rational arithmetic. The re-verification confirmed every closure and also confirmed that the hyperplane and truncation sweeps of §6 are complete as stated.

6. Boundary data, and the CSS case

Nothing in this section is evidence about existence; it is recorded so that the next attempt does not repeat it.

Proposition 6.1. *If S is a $[[14, 3, 5]]$ stabilizer subspace, then for every logical $l \in S^{\perp} \setminus S$, the subspace $T = S + \langle l \rangle$ is a $[[14, 2, \geq 5]]$ stabilizer subspace. Hence every $[[14, 3, 5]]$ is a hyperplane (index-2 subgroup) of some $[[14, 2, 5]]$.*

Proof. T is isotropic of dimension 12 since $l \in S^{\perp}$, and $T^{\perp} \setminus T \subseteq S^{\perp} \setminus S$, so the minimum weight can only go up; the tables' upper bound $d \leq 5$ at $[[14, 2]]$ [Gra] then pins $d(T) = 5$. □

Hyperplane sweeps. All 4,095 hyperplanes of the codetables $[[14, 2, 5]]$ have $d \leq 4$; the same holds for all 4,095 hyperplanes of a second $[[14, 2, 5]]$ found here by transvection annealing (seed 44; stabilizer in `data/seed_1425_anneal44.txt`, verified $d = 5$). So neither known $[[14, 2, 5]]$ sits over a $[[14, 3, 5]]$.

Truncations. All 45 rank-preserving single-qubit truncations of the codetables $[[15, 3, 5]]$ give $d < 5$.

Walks and annealing. A $d \geq 5$-preserving random walk on the $[[14, 2, 5]]$ manifold scores each visited code by its minimum, over nonzero syndromes, of the number of weight-≤ 4 Pauli errors mapping to that syndrome; the value 0 anywhere would yield a candidate $[[14, 3, 5]]$, which an exact re-check would then settle. From the codetables seed, a 15,000,000-step run (seed 101; 4,039,818 accepted states) drove this coverage from 13 to 5 within the first 251,000 steps and never below 5 thereafter; a second seed's runner died early (step $\approx$ 48,000 of the planned 15,000,000) and its partial log is retained. Random-start transvection annealing on the $[[14, 3]]$ objective floors at a single residual weight-4 dual-coset vector at best (restarts typically stall at 1–4). Evidence of a rugged landscape and of a coverage floor on the explored component, not of nonexistence.

Remark 6.2 (the CSS case is Koh et al.'s). Koh, Gong, Diaconu, Tan, Geim, Gullans, Yao, Lukin and Majidy [Koh26] exhaustively enumerate all CSS codes up to $n = 14$ (2.71×10^{10} inequivalent codes) with exact distances valid for impure codes; their Table VI at $(n, k) = (14, 3)$ lists no $d \geq 5$ row and exactly two codes of distance 4. Hence no CSS [[14, 3, 5]] exists, pure or impure, and (distances being invariant under local Cliffords) neither does any local-Clifford image of a CSS code with these parameters. This settles the entire CSS branch of the question by citation, and it subsumes the only CSS statement we had derived independently — that a *pure* CSS [[14, 3, 5]] is impossible because the best $[14, 7]_2$ distance is exactly 4 while $d(C_1) \geq 5$ and $d(C_2^\perp) \geq 5$ would force $\dim C_1 \leq 6$ and $\dim C_1 \geq 11$ [Gra]. That two-line argument survives only as an independent hand proof of two rows of their computed table; the result is theirs. Their catalogue is CSS-only; the open territory for [[14, 3, 5]] is therefore genuinely non-CSS.

Remark 6.3 (partial results toward orders 2, 3, 5). Outside the adversarially re-verified perimeter of Theorem 3.1, the working record contains a partial closure of the order-5 classes by the same machinery: all order-5 monomial elements reduce to pure permutations of cycle type $(5, 1^9)$ or $(5, 5, 1^4)$; the invariant dimensions satisfy $d_0 + 4e = 11$; the branches $(d_0, e) = (11, 0)$ and $(3, 2)$ are impossible — $(11, 0)$ by a weight-2 vector surviving in the field component, $(3, 2)$ for the one-5-cycle type via a second exact-rational LP certificate (`lemmaD_certificate.py`: every additive $(9, 2^{15})$ code has at least 1,471 nonzero vectors of weight ≤ 4) and for the two-5-cycle type by a fixed-space dimension count — leaving only $(d_0, e) = (7, 1)$ open. We record this as work in progress with its certificate, not as part of the theorem.

7. Artifacts and replay

The artifacts sit in `solve/problem-3/` beside the working notes: generators `gen_cyclic.py`, `gen_qc7.py`, `gen_c11.py`, `gen_generic.py`; the distance checker `check1435.c` (single C file: exact check, batch screening, annealing, walks); the independent verifier `verify_1435.py`; `gen_hyper.py`, `truncate15.py`, `order7fixed_branches.py`; reference matrices in `data/` (retrieved from [Gra] on August 5, 2026); and the run outputs behind Table 1 and §4 in `certificates/`. (The hyperplane and truncation sweeps of §6 were confirmed in the adversarial re-verification, but their console outputs were not archived; the empty `c13` class leaves no output file. The directory also retains `sample_c7fixed.py`, the defective sampler disclosed in the Acknowledgments, unhashed; nothing depends on it.) The differentiator is deliberate and worth one plain sentence: the closure certificates — both LP lemmas, the branch enumeration, and the independent distance verifier — replay with the stock Python interpreter and nothing else; no package, no solver, no proof assistant. To replay:

```
cd solve/problem-3
python3 certificates/lemmaC_certificate.py   # Lemma C, exact rationals
python3 certificates/lemmaD_certificate.py   # Lemma D, exact rationals
python3 order7fixed_branches.py              # d0 = 8 branch counts
python3 verify_1435.py data/ct_14_3_stab.txt # control: d=4, "FAIL" by design
cc -O2 -o check1435 check1435.c              # distance checker
python3 gen_cyclic.py | ./check1435 batch 11         # 1,260 cand., 0 hits
python3 gen_qc7.py 1  | ./check1435 batch 11         # QC-7 slice dp=1
python3 gen_c11.py    | ./check1435 batch 11         # 8,415 cand., 0 hits
python3 gen_generic.py shift28 | ./check1435 batch 11  # needs sympy
```

The four `gen_qc7.py` slices (0..3) total 1,298,700 candidates; `gen_generic.py` (the only script with a dependency: SymPy, for one factorization) accepts every twisted class name of Table 1. Expected terminal lines are the `SUMMARY candidates=... hits=0 bad=0` records reproduced in the certificate files; acceptance is by the `SUMMARY` line itself — `candidates` equal to the class count, `hits=0`, `bad=0` — not by exit status (`batch` exits 0 even when `bad>0`). Two expected oddities of a verbatim replay: the independent verifier exits 1 on its control by design (it reports the exact distance 4, then `FAIL: distance 4 < 5`); and the second section of `order7fixed_branches.py` is a superseded block-test attack on the $d_0 = 5$ branch, retained for the record — its closing line "branch NOT closed by this test" is expected, that branch being closed by Lemma 4.2. SHA-256 hashes of every file in `certificates/` (repeated hashes are genuinely identical outputs of independent runs):

```
10bdfe41bb4bc1b2d46e3649d6952fdc8da17c2631cad08e889164ef736488cc  anneal2_s11.out
029aae2eb7cb54809488669f4198b56bfc7796743089b8f49bdb2efffa7db6d4  anneal2_s22.out
7b235c01f0c8a68a17f2478b7745c11913300ea999a913b21d10d14dd4ae4098  anneal2_s33.out
71780fe55a5f3e50927cf3efe4bff13f76c88c55f62620ce5d5a1d50b51ce9bc  anneal2_s44.out
3a49de35dc9175e0b0efe66e593da006d1a83d3764255bc13a96d4ae2d7744ac  anneal_s55.out
d99197bb63d8acfc926c687e05688fdfa012842a91692cc1018b8d8d9fbdd407  c11_result.txt
c4c4abab60c19de617c980c132013cd289ca2933b34e5b1d197dec757c3112bc  cyclic_candidates_1260.txt
517cce87f8c4a3642afe82e8be4196fa4628734a6a8ed0c2e08ee3ff1c715f62  cyclic_result.txt
c0305d1a9307bb8b6ae7d9328c6ea5b62498125e0098a9dfa1217c3c55705e46  lemmaC_certificate.py
7ef4f10dbf47e132957814bc16fd10f71775e64f5e383fad6e98f682036c1492  lemmaD_certificate.py
f3fe027b7fe84f59d70b9fa0e21b11ca667b75d2913011bc1aed9d3c8b4e0d17  order7fixed_full.txt
b81a7565ac6b6be87616917c1ded8f80221efde2b517d03d5e02532cab14c982  order7fixed_summary.txt
abc5f982aff4ce5e7f6aef7548daf2eb6610eda19142b821ec3ffbc96c353b12  qc7_1rho.log
dda0a0d23bf6df85f929a070415f1aedeafc7bbc4af73238554ef69310a7427b  qc7_1rho_result.txt
b7b7ab2b92e8870be0cd184b21aa9b89727cc5f62c7304b3cfc382a29aaf4ca5  qc7_1swap.log
8257c009248694646c5a81daa77fb3bb474316f9b466339b49e240e389626932  qc7_1swap_result.txt
58b70180f2e45b8d55c42218f4f11dd57e38211064c3019d7f2f3534e1447445  qc7_dp0.log
b3220909a34b785cb058e62610a7a9538dd7cb7b8b30b5e4cb9ca3c27d0fbd05  qc7_dp0.out
a11bea4de8c1ff90a25f2844f92158bd135ac935a7dba32acad7d83dd2011d1f  qc7_dp1.log
7e5fc32bb0d4f480fc2d111eb8910d8b204963c7cb7409177100e11a151355ab  qc7_dp1.out
b95ee11c013c340554e6a9791f3b842ffde2bb77b0f9d99a1c373e6e5e43f814  qc7_dp2.log
7e5fc32bb0d4f480fc2d111eb8910d8b204963c7cb7409177100e11a151355ab  qc7_dp2.out
043be1db5d5de02a7d4f2ac12306010a5b8b8ba23229b8223030d9404419bae1  qc7_dp3.log
b3220909a34b785cb058e62610a7a9538dd7cb7b8b30b5e4cb9ca3c27d0fbd05  qc7_dp3.out
393a15714494f641badc0778b4abffa00f1739b6b97f4d077ea52f0601640ca0  qc7_rho_rho2.log
dda0a0d23bf6df85f929a070415f1aedeafc7bbc4af73238554ef69310a7427b  qc7_rho_rho2_result.txt
eded10efb983e8f44a324d731f08b310df7c4f3310a4fc962e7f6f285d9dddac  qc7_swap_rho.log
dda0a0d23bf6df85f929a070415f1aedeafc7bbc4af73238554ef69310a7427b  qc7_swap_rho_result.txt
051d452751f6acb4dc50b496fb919eb3e2f843e1a52fd20f955ab503cef657ed  qc7_swap_s2.log
3feeaba0ed1dc6615dc8a234aaff3ad090582e644a9f6ae2a5de832a2b89bd8b  qc7_swap_s2_result.txt
1fe84310ee3ca4fbd289f1714c72bdb9d3041d0963e553d36fdec4615759d96e  qc7_xcheck.out
c133da9f6f78dcb64204ff6d5f001059521df14a1079d9655f130f7077180c3a  qc7s3.log
dda0a0d23bf6df85f929a070415f1aedeafc7bbc4af73238554ef69310a7427b  qc7s3_result.txt
aae109906f4f8efa3535a441ee67e08823a6bd7cd12c1b1270cac54dad853e97  qc7swap.log
3feeaba0ed1dc6615dc8a234aaff3ad090582e644a9f6ae2a5de832a2b89bd8b  qc7swap_result.txt
263c7234e893f016bd33ead2a8354b575575c15bb2f75872baf74114bdefc4ed  s3shift2.log
dda0a0d23bf6df85f929a070415f1aedeafc7bbc4af73238554ef69310a7427b  s3shift2_result.txt
316b8bd39998c6851ee4819c5a5ad45d5f8fb722d697290c1552daebbc24ee9e  s3shift_result.txt
93a7314da75f297284e912578c594bb14dbfc6182479aaa683ae087714ad0675  shift28.log
86a4b81844ebc832577ee7221ff8cc2b9ba2a40f8383c1096871974a985995ac  shift28_result.txt
2ac2e1ed208e3d7f29f39059f070681d4d637d911fbc589e7ed0a0a05e7dcf61  swapshift_result.txt
3fe92f126b0808dbf02a6fb5f45381d224f9f998d8251dbd1fd84d77a8358dcf  walk2_s101.out
71a58ebd0c85a2416713da85b7747581ed56cb515b371887682a3bd111cffe15  walk2_s202.out
```

and of the tools and reference data they replay against:

```
95599be7a1ea8f22b13328fef3c9c9e8fb489ef78e1c7d535007db6b736119d5  check1435.c
2527266d13b604029120ed11174730ef1ed3fe6d5405d8fe3f82c21668ae0d3c  verify_1435.py
77ee81af746879b8b56c3e1e259b26fd07c0ccb9be187871a61b08ee3e0762d8  gen_cyclic.py
9e9b4ecd954c848acf841fc1a227f911a2497ee81c7988853545105f9c5c87b7  gen_qc7.py
6713b83cdc18d5fc47e000530953180044658a402728d235cf566d26def50dd7  gen_c11.py
266ed6d4358bf1571de621e0819f4d46fba56d14015524a1e79239a7c9e4903e  gen_generic.py
23f4b2ccc019ec1ca5895f980a4687ee7a93cd35e9b4799c48cd0b97cf214674  gen_hyper.py
3782bda28af7426292f7409ba1971288ed788bbe02882debd7e5ff60fde88f6a  truncate15.py
e6f94ea565cda966bb02a70f919e54e67b4c83081e11feefd11d8aa7b765977d  order7fixed_branches.py
7668dcc98ef70aafb215c908728e19e177107879f683b4ff0b19a0ab104d0ee3  data/ct_14_3_stab.txt
80fc7aef58469c67d52faa3f73562041b051905284118f9404a0059102330505  data/ct_14_2_stab.txt
53c055242bdb7a77bf3c31884927464d89a49a132913a80ce209d887e0e8f7b1  data/ct_15_3_stab.txt
e6fce037fc1f0f3b8535ae30076678ee7c2359d22b5aff16ca402834dc013489  data/seed_1425_anneal44.txt
```

Acknowledgments

The problem is Grassl's tables' and Ball, Centelles and Huber's: the $[[14, 3]]$ entry [Gra] poses the question silently (its $[[14, 3, 4]]$ construction is dated June 30, 2005), and [BCH20] posed it in print. The debt to Koh, Gong, Diaconu, Tan, Geim, Gullans, Yao, Lukin and Majidy [Koh26] is recorded in Remark 6.2: their enumeration closes the entire CSS branch and subsumes the one CSS statement we had derived ourselves. The prescribed-automorphism method is classical and Huffman's account [Huf98] is the one we followed; the GF(4)/symplectic framework and the additive MacWilliams identities underlying Lemma 4.2 are Calderbank, Rains, Shor and Sloane's [CRSS98], and the linear programming method is Delsarte's [Del73]. All computations were carried out with Claude Fable 5 (Anthropic), directed by the author. One sampling-based falsification run present in an earlier working draft was removed after the

adversarial re-verification found its sampler defective; the certified enumerations above do not depend on it.

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